



SUCCESS SKILL: CRITICAL THINKING

BACKGROUND

This rubric is aligned to the [NCIES literature review](#) on critical thinking. The value proposition is a clear research-based definition of critical thinking as a success skill. This research-based rubric¹ is designed to provide useful, formative information that teachers can use to guide instruction and provide feedback to students and students can use to reflect on the quality of their critical thinking. Student performance is described along a continuum and balances content-specific and the more general nature of critical thinking. Rubrics are provided by grade span, K-2, 3-5, and 6-12. The K-2 rubric includes both a teacher version and a version for primary students or early readers. Highlights from the National Center for Assessment's Literature Review are provided below. The research-based definition of critical thinking is:

"...purposeful, self-regulatory judgment which results in interpretation, analysis, evaluation, and inference, as well as explanation of the evidential, conceptual, methodological, criteriological, or contextual considerations upon which that judgment is based....The ideal critical thinker is habitually inquisitive, well-informed, trustful of reason, open-minded, flexible, fair-minded in evaluation, honest in facing personal biases, prudent in making judgments, willing to reconsider...and persistent in seeking results which are as precise as the subject and the circumstances of inquiry permit" (Facione, 1990, p. 3).

BUILDING AND ASSESSING CRITICAL THINKING SKILLS USING CULTURALLY-RELEVANT GOLD STANDARD PBL

- **Use research-based guidance to teach critical thinking.**
 - ***Give students multiple opportunities to demonstrate critical thinking in various subjects throughout the year.*** "The research evidence suggests that while critical thinking is instructionally sensitive, it does not necessarily change over the short term" (Evans, 2020, p. 14). Ideally, a collection of student work overtime may be most appropriate to support a judgment about students' critical thinking. The literature is unclear regarding how much growth in students' critical thinking we should expect to see in a school year or how critical thinking competencies develop over time (Evans, 2020).
 - ***Use metacognition "thinking about thinking" to support critical thinking.*** Teachers who model, scaffold, and support student use of meta-cognitive strategies (e.g., visualizing, making connections, monitoring and clarifying, asking/generating questions) demonstrate a statistically-significant positive impact on student learning ([Hattie, 2009](#)). These strategies can support the development of critical thinking skills.
- **Use Gold Standard PBL to support student's development of critical thinking skills and assess them within a project.**
 - **Critical thinking in the context of Gold Standard PBL addresses multiple research-based assessment criteria.** To assess the quality of student critical thinking, Evans (2020) suggests that tasks need to:
 - ***Prompt complex judgments.*** The design of the project, tasks, and assessments need to be rigorous and purposefully designed for critical thinking.
 - ***Be open-ended.*** Tasks that are open-ended require students to determine what is relevant, plan how they will demonstrate their knowledge and offer multiple pathways for solutions. Open-ended tasks "allow for multiple, defensible solutions and, further, require students to apply (or transfer) their learning to novel situations" (p. 12).
 - ***Be authentic.*** The design features real-world contexts, tasks, tools, or impact. The PBLWorks Blog, [What Does it Take for a Project to be "Authentic"?](#) is a helpful resource for planning.
 - ***Make student thinking visible.*** Offer students opportunities within the work to make their thinking visible to their group: [Structured reflections](#).
 - ***Include structured opportunities to build critical thinking skills.*** A Gold Standard project needs to be intentionally designed to develop complex thinking. This happens through alignment across a project with intentional [scaffolding](#) throughout the [project path](#). Read our [blog](#) to learn more about *A PBL Culture of Thinking: Routines*.
 - ***Reflect local culture and context.*** While [PBL supports culturally responsive teaching practices](#), it is important to note that while assessing the quality of critical thinking, cultural differences must be considered. It is important to review and pilot test in local contexts to ensure it accurately reflects how students across sociocultural contexts and conditions demonstrate critical thinking competencies (Evans, 2020). Teachers should know their own cultural lens and recognize how that may influence the interpretation of culturally and linguistically diverse student learning behaviors ([Hammond, 2013](#)).

¹ These rubrics were created by National Center for Improvement of Educational Assessment (Center for Assessment) in collaboration with PBLWorks. This version of the rubric integrates asset-based language based on the *Ready for Rigor: A Framework for Culturally Responsive Teaching* (Hammond, 2013).



Critical Thinking Rubric Grade K-2 (For Teachers)

Critical thinking occurs when students make judgments demonstrating explanation & analysis (uncovering interrelationships of topics and concepts), inferences & interpretation (making inferences and generalizing knowledge), and evaluation (determining the quality of information and arguments).

DIMENSION	 Quality of Student Thinking			
	<i>Select the column where most of the criteria describe what a student knows and can do independently.</i>			
	BEGINNING	EMERGING	DEVELOPING	DEMONSTRATING
Explanation/Analysis	Explains using unrelated examples or opinions.	Explains using a combination of related and unrelated examples or opinions.	Explains using important facts and/or examples.	Explains using important facts and/or evidence, categorizing by obvious similarities and differences.
Interpretation/ Inference	Makes predictions unrelated to the topic.	Makes predictions about the topic related to personal experience.	Makes predictions about the topic using personal experience AND new content as support.	- Makes plausible predictions about the topic using prior content knowledge AND new information as support. - Uses information to draw a simple conclusion.
Evaluation	Identifies all information as equally trustworthy.	Identifies the difference between facts and opinions.	Tells which facts and/or opinions are more trustworthy using personal experience.	Selects which facts and/or opinions are more trustworthy using provided criteria.



EXAMPLE: Critical Thinking Rubric Grade K-2 (For Students)

DIMENSION	 My Thinking			
	<i>Show what you know and can do independently.</i>			
	BEGINNING	EMERGING	DEVELOPING	DEMONSTRATING
How do I explain and analyze?	• I use unrelated examples or opinions	• I use some related and some unrelated examples or opinions.	• I use important facts and/or examples.	• I use important facts and evidence. • I group by similarities and differences.
How do I make inferences and conclusions?	• I make predictions, but they are unrelated to the topic.	• I make predictions about the topic using what I already know.	• I make predictions about the topic using what I already know AND new information from our learning materials.	• I make predictions about the topic using what I already know AND new information from our learning materials. • My prediction is realistic for the topic. • I use information to draw a conclusion.
How do I determine the quality of information and arguments?	• I see all information as trustworthy.	• I tell the difference between facts and opinions.	• I use my personal experience to tell which facts and/or opinions are more trustworthy.	• I use rules I am given to select which facts and/or opinions are more trustworthy.